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AN ILLUSTRATED TAXONOMY OF METADATA

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TABLE OF CONTENTS

RECORD LEVEL TECHNICAL METADATA	3
STRUCTURES OF TECHNICAL METADATA.....	13
BUSINESS METADATA	15

*For years there has been the definition of metadata -
"metadata is data about data."*

Today, this definition is almost meaningless. It neither describes what metadata is nor does it provide any sort of elucidation about metadata. It leaves the reader clueless about what actually constitutes metadata.

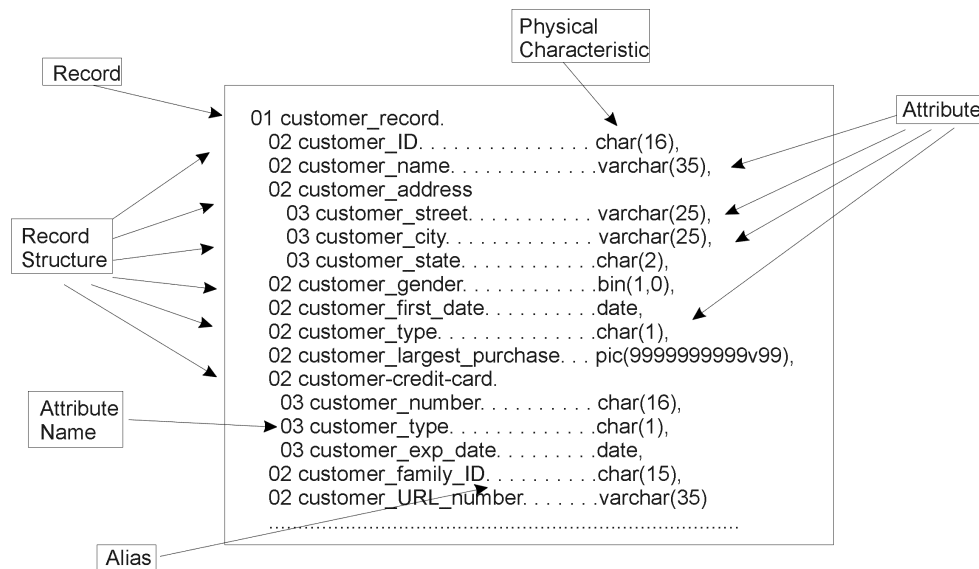
One of the difficulties of defining metadata is that it has such a broad set of meanings. Literally, metadata can be taken to mean any form or description of data other than the actual data itself. This includes a whole lot of territory, as shall be seen in this white paper.

To understand what metadata really is, it is necessary to go much further than this simplistic definition that really describes nothing. An effective and useful way to describe metadata is through a taxonomy of metadata in which the common visceral components of metadata are identified, as well as how those components relate to each other.

In general, metadata is divided into two separate types: technical metadata and business metadata. Within the category of technical metadata there are many subdivisions: record level data, structure data, operating data, and so forth.

RECORD LEVEL TECHNICAL METADATA

Record level technical metadata exists to describe the common records of data that are used to store structured data. Figure 1 depicts a record with its attendant metadata.



Some Common Types Of Technical Metadata

Figure 1

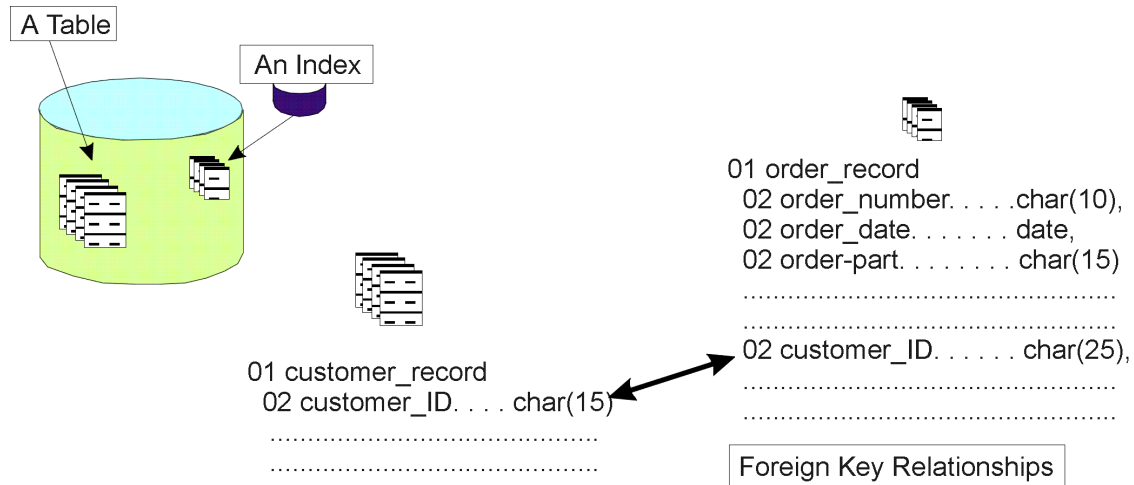
At the record level there is the:

- **Record** - the record is the unit of data that encompasses one complete unit of metadata. The record may be for a transaction, such as an order, or the record may be about a customer, where a snapshot is taken at one moment in time. The

record is the most basic unit of data storage. A record is sometimes called a tuple or a row, depending on the DBMS and the context of usage.

- Record structure - the record structure refers to the organization of data inside the record. The internal organization usually specifies what attributes the record will have, the order in which the attributes will appear, the physical characteristics of the attributes, whether the appearance of an attribute is optional or not, and so forth.
- Physical characteristic - the physical characteristic of an attribute is the specification of what physical form the attribute can take. The size of the form and the nature of the form is specified. The nature of the form refers to the physical interpretation of the form, such as CHAR, VARCHAR, BIN, PIC, DEC FIXED, DATE, and so forth. The size of the form refers to how many bytes will be dedicated to the attribute.
- Attribute - an attribute is the most basic unit of data in a record. An attribute contains data about one type of information, such as CUSTOMER_NAME, CUSTOMER_STREET_ADDRESS, and so forth.
- Attribute name - the commonly held identification of the attribute. Each attribute has a name. The name is used by the programmer for internal program manipulation or by the end user for reference to specific parts of the record. As a rule, the more descriptive the name the better.
- Attribute alias - an alternative attribute name.
- Table - a collection of records all of the same type. Also known as a database.
- Table name - the name the table or database goes by.
- Index - an index is a grouping of attributes or representations of attributes external to the record used to quickly find one or more records. An index becomes important to the way the data base is designed because it allows data to be accessed efficiently. But if a data base record has too many indexes then there is the issue of the overhead of creating and maintaining the index.
- Foreign key relationship - one record is related to another by means of a foreign key relationship. A foreign key relationship consists of a key in one type of record and a foreign key in another type of record. By means of matching the values in the record, a foreign key relationship is created. A foreign key relationship then is a means of relating two different record types by a common value.
- Recursive relationship - a recursive relationship is one where a foreign key relationship exists between different records of the same type. For example, there may be a data base record about an individual. One record is for John Mayberry. Another record is for Mary Mayberry. There is a relationship between the record for John and the record for Mary because John is married to Mary. This is a form of a recursive relationship.

Figure 2 illustrates an index and foreign key relationship.



More Types Of Metadata

Figure 2

Figure 3 shows some other forms of metadata.

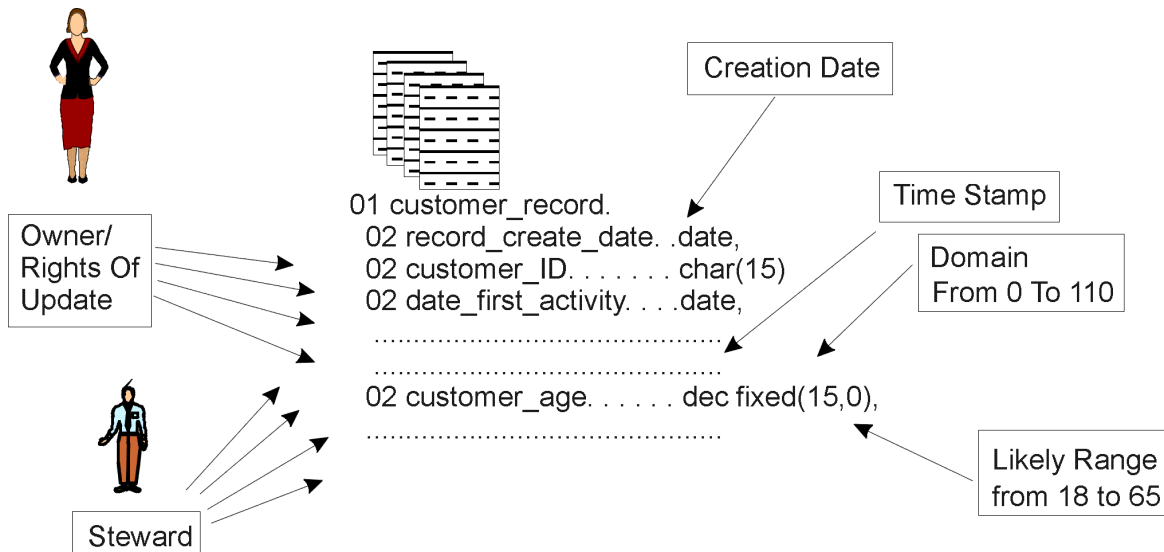


Figure 3

There are several types of attributes within the record:

- Creation date - creation date refers to the moment in time when a record was first formed.
- Time stamp - time stamp is the moment in time when a record was entered into a table. Note that time stamp may or may not be the same thing as creation date.

- Attribute domain - an attribute domain is the set of values that an attribute may have. For example the attribute GENDER may have a value of M or F. Or the attribute AMOUNT_OF_SALE may have a value of anything numeric.
- Likely range - the likely range of an attribute is the likely set of values. For example, for AGE_OF_AUTO_INSUREE, the age may be from 18 years of age to 65 years of age. It is unlikely that the insurance company will insure anyone at the age of 6 or over the age of 100.
- Owner/rights of update - the owner of a record is someone who has been designated as the responsible party for the content of data. If it is discovered that John Wood's birth date is July 20, 1950, when his record shows that his birth date is April 13, 1952, then it is the responsibility of the owner to correct the data inside the record.
- Steward - the steward of data is responsible not for the accuracy of the contents of the data but for the physical safety and integrity of the data. For example, a data base goes down. The steward is responsible for recovering and restoring the data base to a state of usability.

Figure 4 shows that a record may have key values.

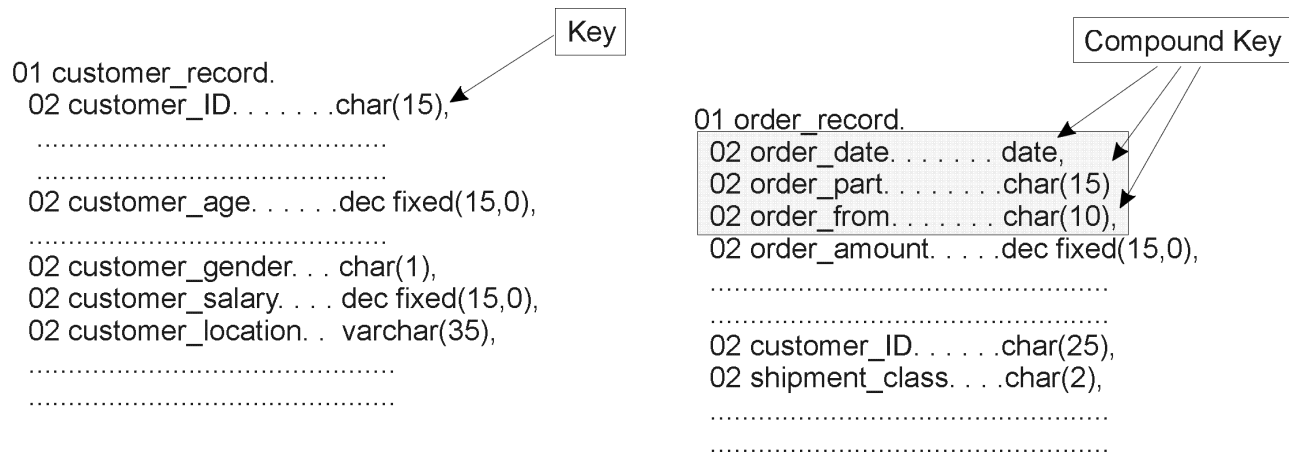


Figure 4

- Key - the key is an attribute that can be used for quick identification of a record or records. For example a key may be the social security number for an individual - SSNO or the gender of a person - GENDER. A key may be unique or non-unique. A key may be made up of multiple attributes combined together to form a compound key. A record may or may not have a key. In addition, a record may have multiple keys.
- Unique key - a unique key is a key that singularly identifies a record. For example for an individual's record that is identified by social security number, there had better not be duplicate keys for different records. (The implication is that two people would have the same number.)

- Non-unique key - a non-unique key is one that identifies a record but where more than one record can be identified. For example, a car manufacturer may have a data base for all owners of a Honda.
- Compound key - a compound key is a key that is made up of multiple attributes.

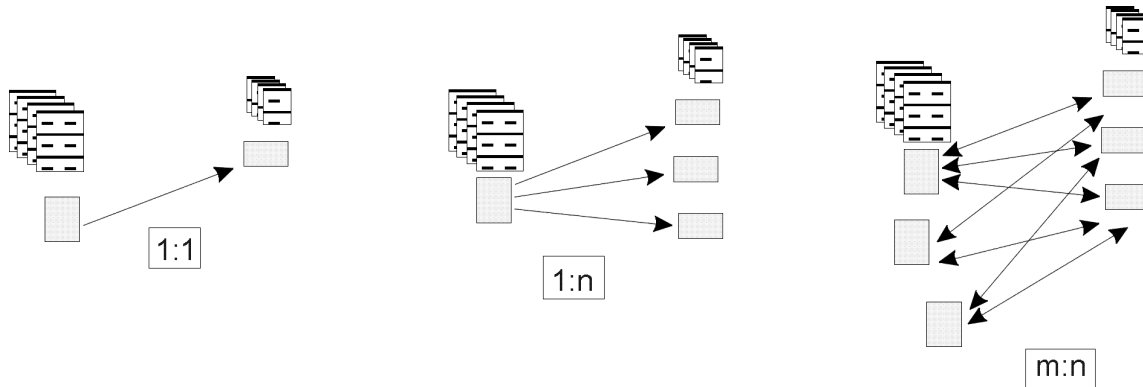


Figure 5

Figure 5 shows that relationships such as foreign key relationships can have different degrees of cardinality. The degrees of cardinality refer to the relationship of the key to the number of foreign keys that can exist.

- 1:1 relationship - a 1:1 relationship is one where there is exactly one key and one foreign key. These relationships exist but are extremely rare.
- 1:n relationship - a 1:n relationship is one where a key may have from zero to many foreign keys. For example, the key may be an owner of a car and the foreign key may be the car that is owned. An owner may own zero, one, two or more cars. A 1:n relationship is very common.
- m:n relationship - an m:n relationship is one where a key may have multiple foreign keys and a foreign key may have multiple keys. Such relationships are very common. Consider the ownership of a car. A person may own multiple cars, and a car may be owned (or have been owned) by multiple owners.

Figure 6 shows that attributes in a record can have a definition.

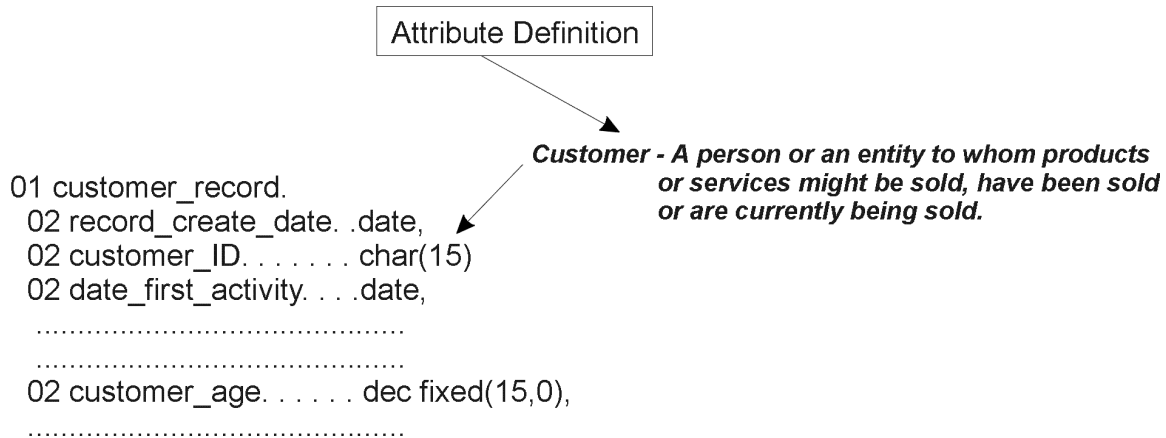


Figure 6

- Attribute definition - an attribute definition is the description of the unique characteristics of an attribute

Figure 7 shows that tables also can have a definition.

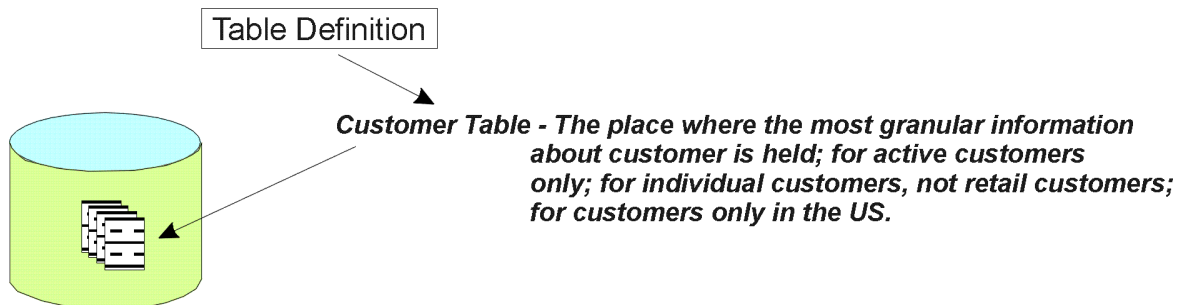


Figure 7

- Table definition - a table definition is the description of the unique characteristics of a table

Record level metadata is not the only type of metadata there is. Another important type is that associated with the model of the data found in the corporate environment. This model is an abstraction of the structure and content of the data and it takes many forms. The most popular forms will be discussed here as common components of metadata.



- Figure 9 shows one of the basic building blocks of the data model.



- Entity - a major subject area. Subject areas are non-overlapping.

Figure 10 shows that the entity is "exploded" into a mid level data model.

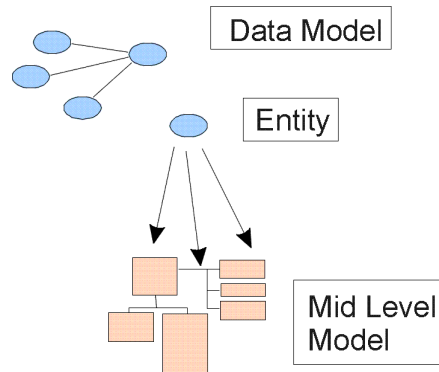


Figure 10

In the mid level data model, the structure and the content of the entity is identified and refined.

Figure 11 shows that the physical data model is created from the mid level data model.

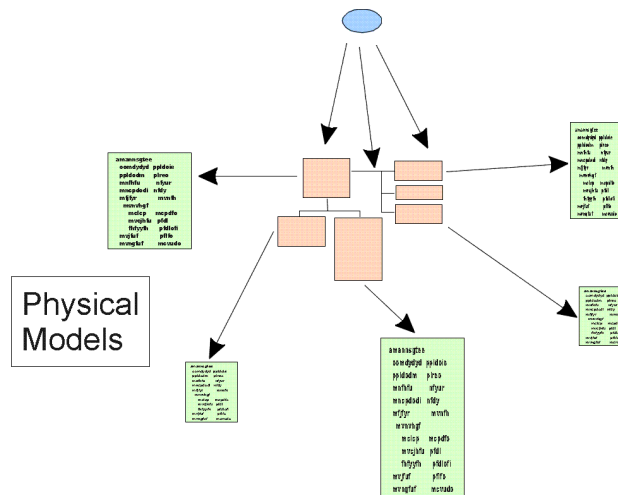


Figure 11

Figure 12 illustrates that each entity or major subject area has its own definition.

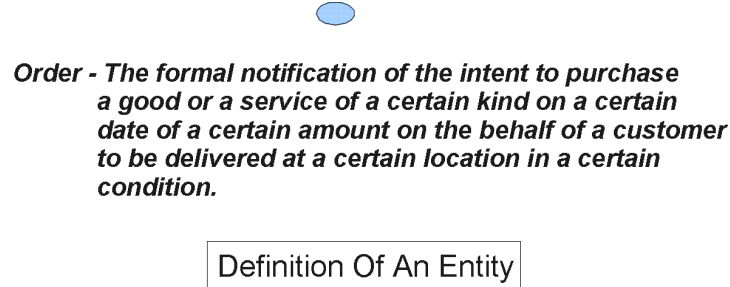


Figure 12

Figure 13 denotes where the relationships are found in the ERD.

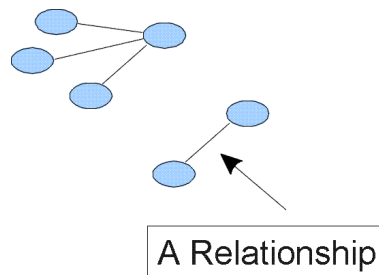


Figure 13

While record level metadata is important and modeling metadata is also important, there is yet another level of technical metadata. That level of metadata is the physical measurement of metadata. Figure 14 depicts some of the important physical measurements of metadata.

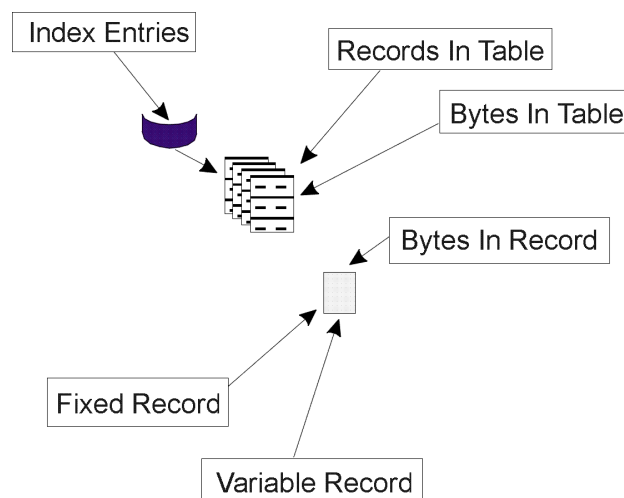


Figure 14

Some of the physical measurements of metadata are:

- Index entries - when an index is created, it may have any number of occurrences that are found inside of it. The type of index that is created, the table that is being pointed to, and how the designer has structured the index all determine the number of index entries.
- Records in table - the number of records in a table refers to the number of occurrences of a record.
- Bytes in table - the number of bytes in a table refers to the total bytes needed to hold the data. Note that the number of records times the bytes per record is not the same thing as the number of bytes needed for a table.
- Bytes in record - the bytes in record is a measurement of how much space on the average that a record takes. The number can be fixed or variable. If the number is variable then the average byte per record becomes meaningful.
- Fixed record - a fixed record is one whose number of bytes is the same for all records.
- Variable record - a variable record is one whose number of bytes may vary from one record to another.

Figure 15 shows the growth rate of a table.

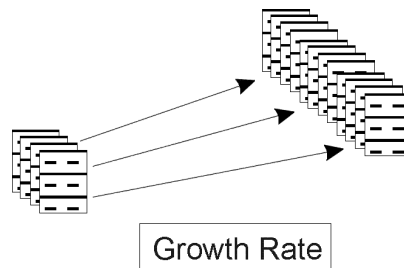


Figure 15

Tables grow as data is added. Of particular interest is the rate of growth.

- Rate of table growth - the rate of table growth is calculated by adding the new occurrences of data and subtracting those that are deleted or are archived off.

Another level of technical metadata is that of architectural metadata. Architectural metadata refers to the type of structure that the data is housed in, and how that structure fits with other structures. Figure 16 shows some basic architecture metadata features.

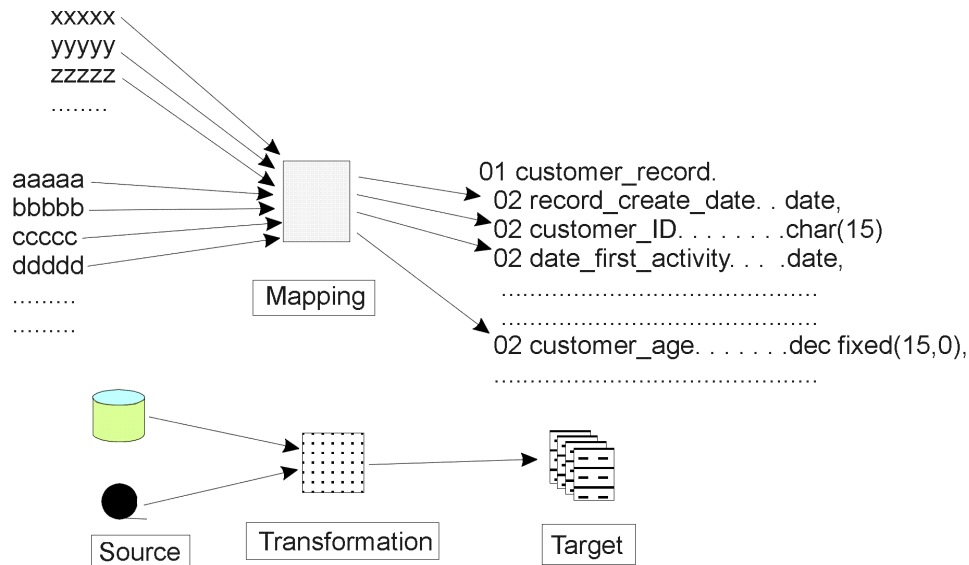


Figure 16

- Mapping - mapping refers to the identification of the source of data. Data is moved from one part of the environment to another. The source is the place where data originates. Mapping can include simple movement, summarization, aggregation, merging, and so forth.
- Source - the source of data is the structure that houses the data before being moved.
- Transformation - transformation refers to the conversion, cleansing, and integration of data as it passes from a source to a target.
- Target - a target is the place where the data lands after passing through the transformation process.

STRUCTURES OF TECHNICAL METADATA

There are different kinds of structures found throughout the architecture, including:

- Legacy environment - the legacy application environment is the place where detailed transactions are done. These applications for the most part were written years ago. They have been maintained to the point where they are threadbare. They often represent business requirements that are years old. The data that is collected here is a by-product of the transactions that have been collected. Generally, the applications found here are unintegrated with any other applications. Each application is its own standalone environment.
- ERP applications - ERP applications are those that are provided by an ERP vendor. The ERP application is integrated, at least within the confines of the ERP environment. Transactions execute inside the ERP environment. The data found in the ERP environment is a by-product of the transactions that have been run.
- Staging area - the staging area is the place where data is deposited after it leaves the application. The staging area represents a place where data can be gathered with no performance impact on the application. Staging areas are usually needed only where there is a large amount of processing or where there is a large

amount of data to be processed. The staging area is not normally accessible to any other processing other than that of collection of data from an application and preparation for entry into the ETL environment. No reporting or other processing is done in the staging area.

- ETL - the ETL (extract/transformation/load) layer of processing is where the unintegrated and uncleansed data coming from the different applications is processed. The purpose of the ETL layer is to integrate and cleanse data in preparation for entry into the data warehouse and/or the corporate ODS. ETL processing does at least the following:
 - o converts data,
 - o reformats data,
 - o merges data,
 - o restructures data,
 - o alters key values,
 - o supplies default values if needed,
 - o resequences data,
 - o cleanses data,
 - o verifies domain ranges of data,
 - o verifies inter attribute relationships,
 - o alters DBMS technologies if needed,
 - o alters operating systems if needed,
 - o summarizes data if needed,
 - o separates data if needed,
 - o combines attributes if needed,
 - o brings in data from other sources if needed, and so forth.
- Enterprise data warehouse - the enterprise data warehouse is the place where granular data resides. The data found in the enterprise data warehouse is data that is detailed, historical, and integrated. Data enters the data warehouse after having passed through the ETL layer.
- Corporate Information Factory - the data warehouse is at the center of the Corporate Information Factory. All other analytical components feed off of the data warehouse.
- ODS - the corporate Operational Data Store (ODS) is a hybrid structure where some operational aspects are found and some DSS, integrated aspects are found. The corporate ODS supports online transaction processing (2/3 second response time) and direct update processing, something not supported in the data warehouse. But the corporate ODS supports DSS processing using integrated data, just like the data warehouse. The ODS is fed by the ETL layer. In turn, the ODS feeds the data warehouse as its data ages.
- Data mart - the data mart is the place where different departments place their own data for DSS processing. The data mart is unique to a department or even a sub department. Typically the departments that have their own data mart are the finance, sales, marketing, accounting, and human resource departments. The granular data found in the data warehouse is gathered and reshaped into an optimal form for the department. Usually, there is much less data in a data mart than in a data warehouse because data mart data is summarized and

denormalized, and because the data mart contains much less historical data. The data structure typical of the data mart environment is the star join, where there is a fact table and dimension tables. Multi dimensional technology is optimal for the data mart environment.

- DSS applications - DSS applications are the place where a single function is accomplished. The function usually traverses across different departments. Typical DSS applications are CRM, elasticity analysis, churn management, and so forth.
- Exploration warehouse - heavy statistical processing occurs in the exploration/data mining facility. The separation of the exploration/data mining facility from the rest of the data warehouse environment means that the processing that occurs in the exploration/data mining facility has no performance impact on the other processing that occurs in the data warehouse environment.
- Near line/alternate storage - near line, or alternative storage, is different from classical disk storage. There is so much data in a data warehouse environment that some form of overflow storage becomes mandatory. In addition, the volume in a mature data warehouse is so large that data starts to separate into two classes - actively used storage and inactively used storage (or dormant data). As time passes and the data warehouse grows, the amount of dormant data greatly surpasses the amount of actively used data. At that point it does not make sense technologically or economically to continue to place all data on high-performance disk storage. Near line or alternative storage presents an attractive alternative to disk storage, both in lowering the cost of storage and, ironically, in enhancing performance. Of course, disk vendors always take the perspective that near line storage hurts performance. Such is the case in online processing where the probability of access of any given unit of data is the same. But in the world of large data warehouses, where vast sectors of data have a very low probability of access, the inclusion of all data onto disk storage actually hurts overall performance.
- Oper mart - an oper mart is a structure that does DSS processing on top of an ODS. An oper mart is very similar to a data mart with one major exception. In a data mart the data that is fed into the data mart is stable. The data mart is usually time delimited so that if the data needs to be reloaded, it is simply a matter of specifying the correct dates. But an oper mart sits on top of an ODS and an ODS can be directly updated. This means that an oper mart can take data from the ODS, do analysis and make a report. But if the oper mart returns to the ODS for a refreshment of data even a minute later, then the ODS may have been updated and the same data that was loaded a minute earlier is not available to the oper mart. Therefore, an oper mart operates on the basis of analysis of ever changing data, while the data mart operates on a very stable and solid foundation of data.

BUSINESS METADATA

Business metadata is similar to technical metadata except that business metadata applies strictly to the businessperson. The businessperson is able to read and understand business metadata in language that is comprehensible. Business metadata provides context for explaining business information.

From a content standpoint there is significant overlap between technical metadata and business metadata. Typical elements of business metadata include:

- subject areas
- definitions
- relationships
- formula
- alias
- common terms
- sources
- transformations
- targets
- subsets of
- selection criteria
- date of calculation
- calculated by
- owner
- and so forth.

IN SUMMARY

The preceding discussions have outlined the most important aspects of metadata content. Metadata can be technical or business. Technical metadata may relate to records, measurements, architecture, or structure. Business metadata is similar to technical metadata except that business metadata applies strictly to the businessperson and provides context for explaining business information.